



US012410834B2

(12) **United States Patent**
Zhao et al.

(10) **Patent No.:** **US 12,410,834 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **LINEAR MOTION GUIDE APPARATUS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 121 days.

(21) Appl. No.: **18/263,298**

(22) PCT Filed: **Nov. 5, 2021**

(86) PCT No.: **PCT/JP2021/040692**

§ 371 (c)(1),

(2) Date: **Jul. 27, 2023**

(87) PCT Pub. No.: **WO2022/163051**

PCT Pub. Date: **Aug. 4, 2022**

(65) **Prior Publication Data**

US 2024/0125352 A1 Apr. 18, 2024

(30) **Foreign Application Priority Data**

Jan. 29, 2021 (JP) 2021-012649

(51) **Int. Cl.**

F16C 33/30 (2006.01)

F16C 29/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **F16C 33/306** (2013.01); **F16C 29/005** (2013.01); **F16C 29/041** (2013.01); **F16C 33/46** (2013.01)

(58) **Field of Classification Search**

CPC **F16C 29/04**; **F16C 29/041**; **F16C 29/043**; **F16C 29/045**; **F16C 29/046**; **F16C 29/048**; **F16C 33/306**

See application file for complete search history.

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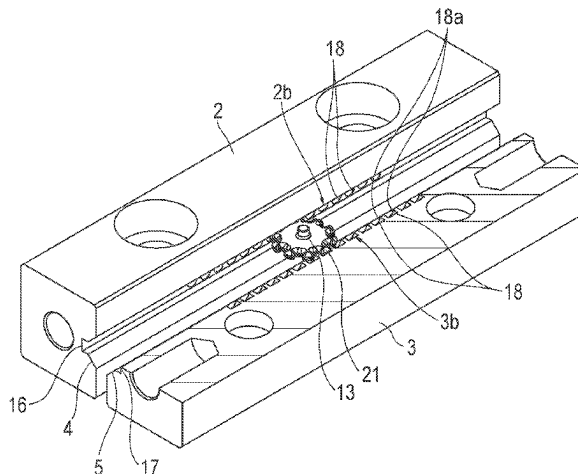
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(57) **ABSTRACT**

A linear motion guide apparatus in which a guide rail can be readily provided with a meshing portion that meshes with a gear is provided. The linear motion guide apparatus includes: two guide rails that are movable relative to each other in a longitudinal direction thereof via at least one rolling element; and a cage configured to hold the rolling element and rotatably hold a gear engaging with a meshing portion provided to at least one of the two guide rails. The meshing portion includes a plurality of circular holes arranged in the longitudinal direction on the guide rail. Each of the circular holes has a wall surface formed in a cylindrical shape.

12 Claims, 9 Drawing Sheets



- (51) **Int. Cl.**
F16C 29/04 (2006.01)
F16C 33/46 (2006.01)

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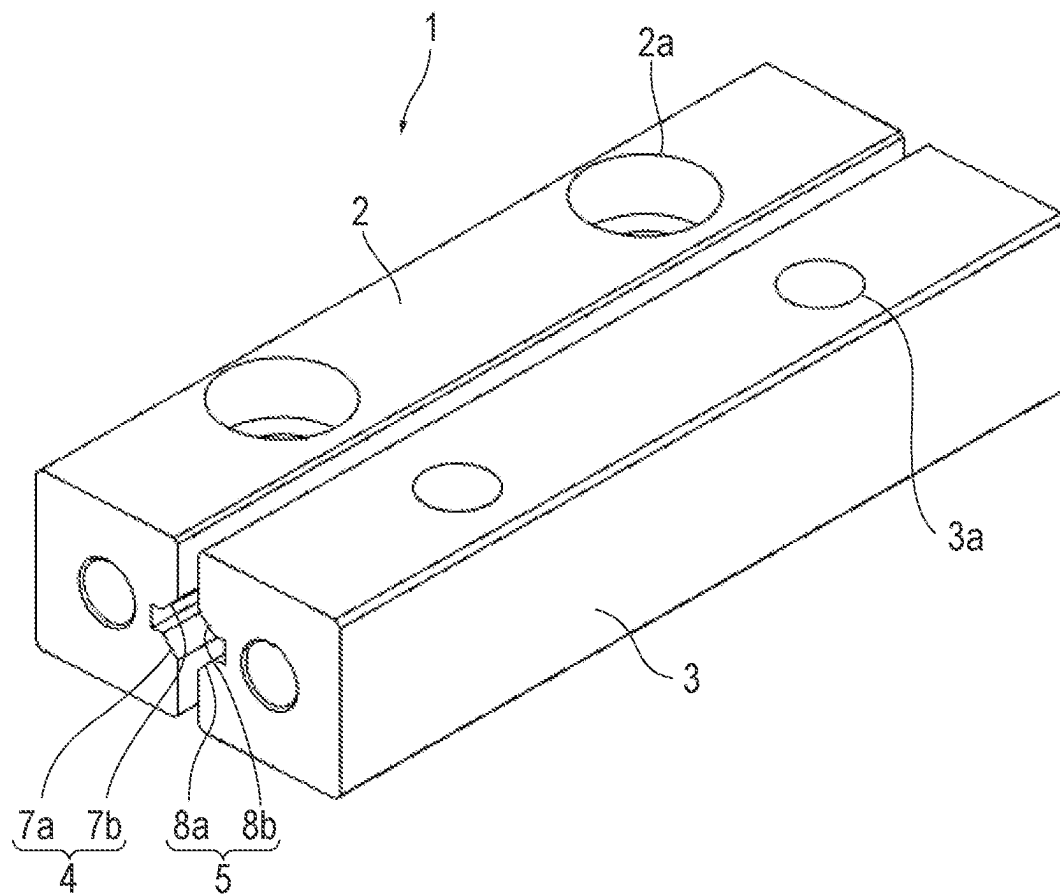
FIG. 1

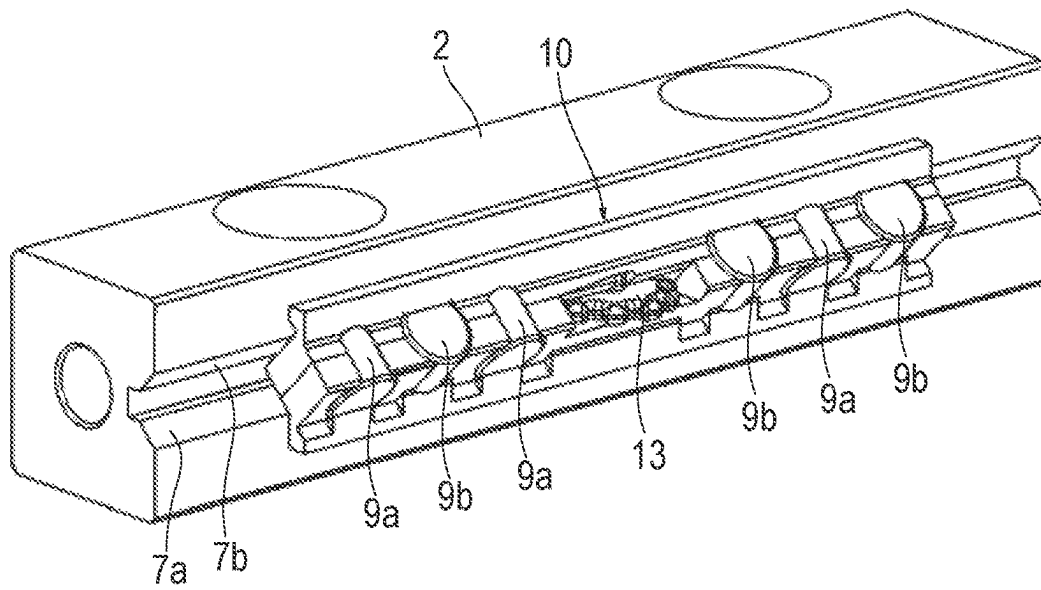
FIG. 2

FIG. 3

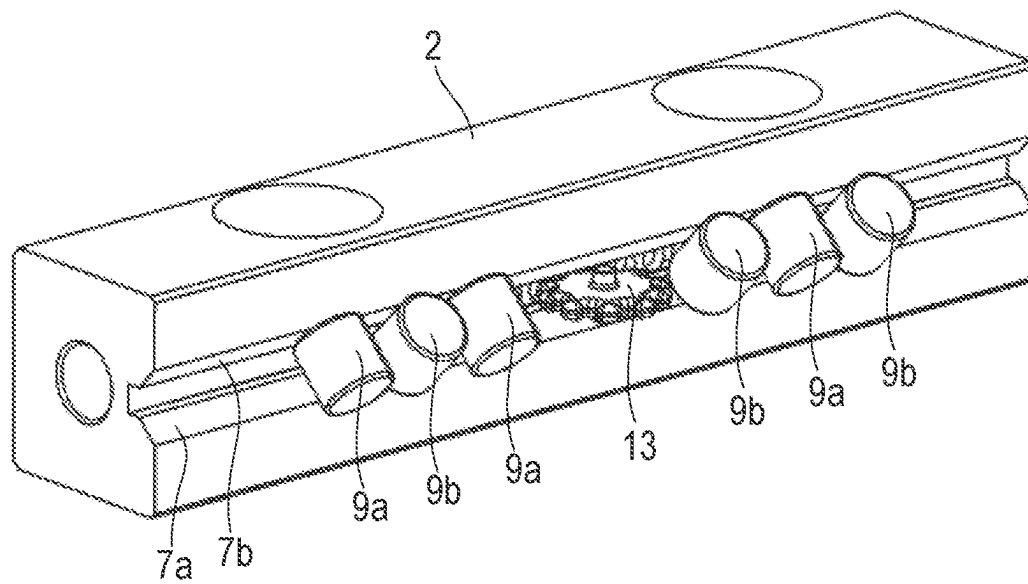


FIG. 4

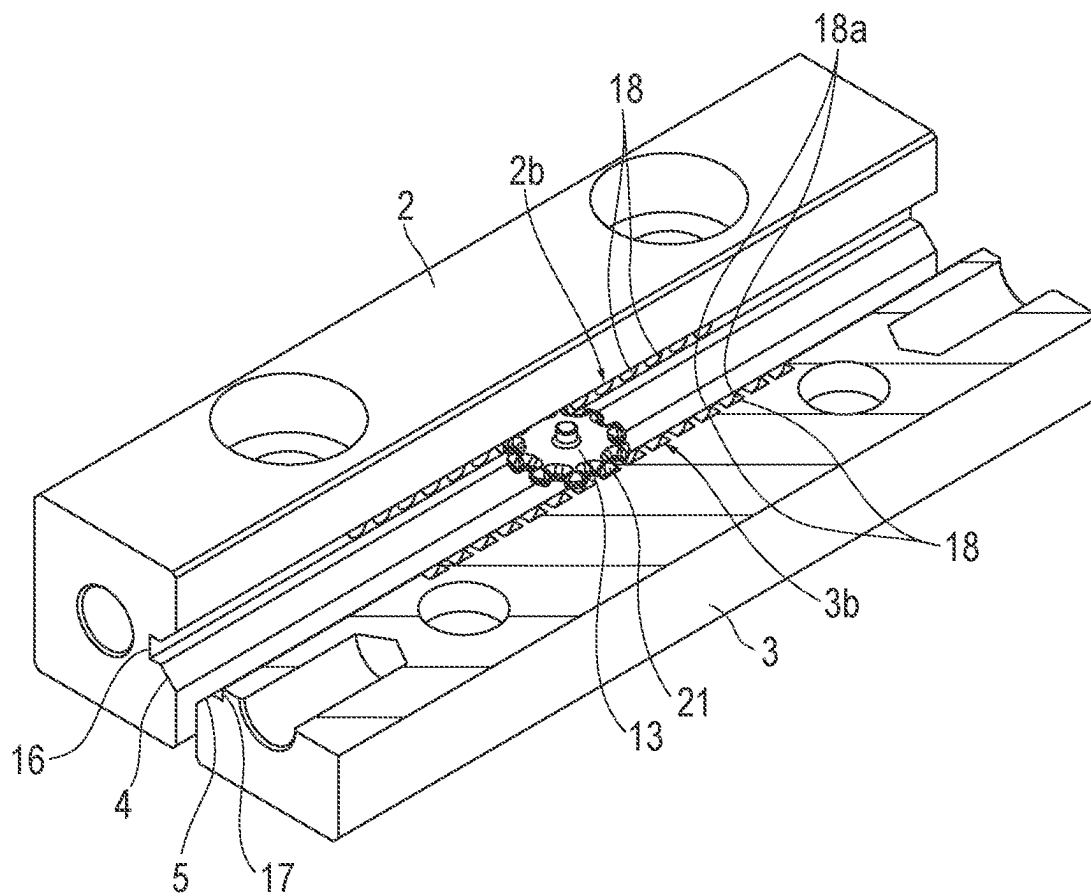


FIG. 5A

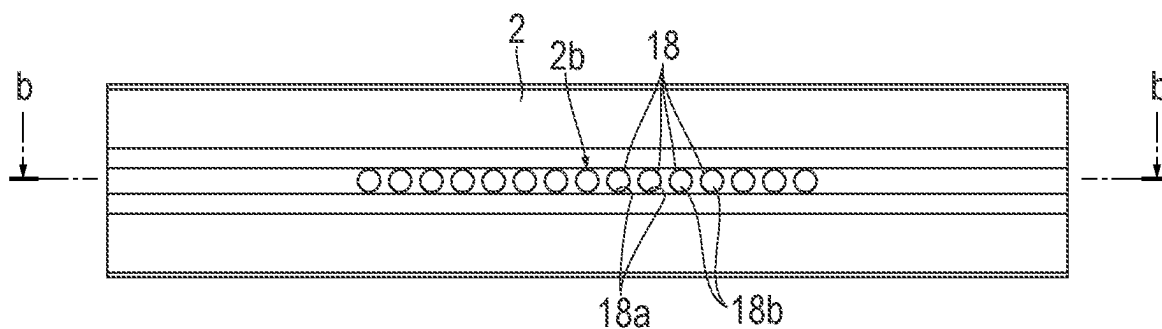


FIG. 5B

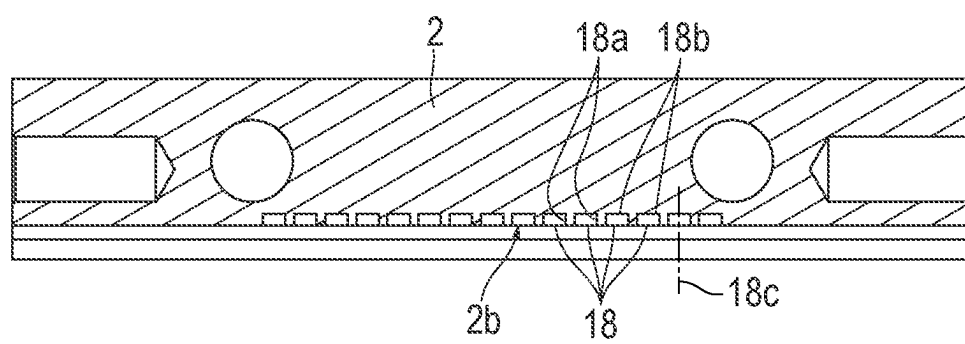


FIG. 6A

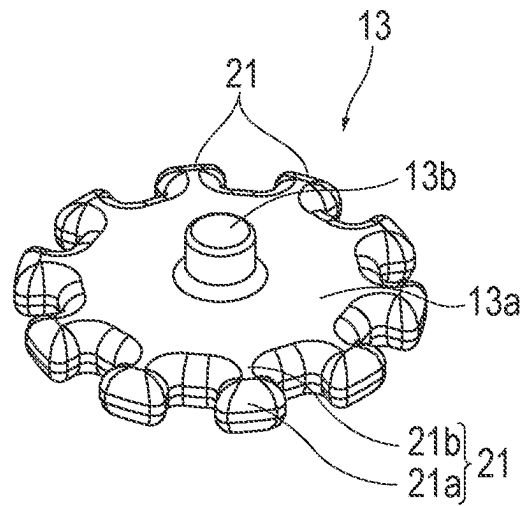


FIG. 6B

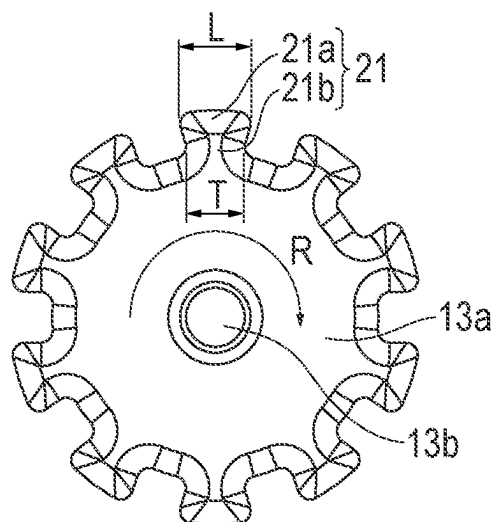


FIG. 6C

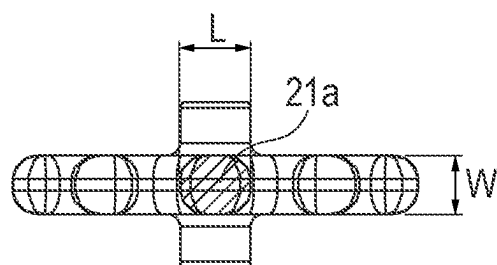


FIG. 7A

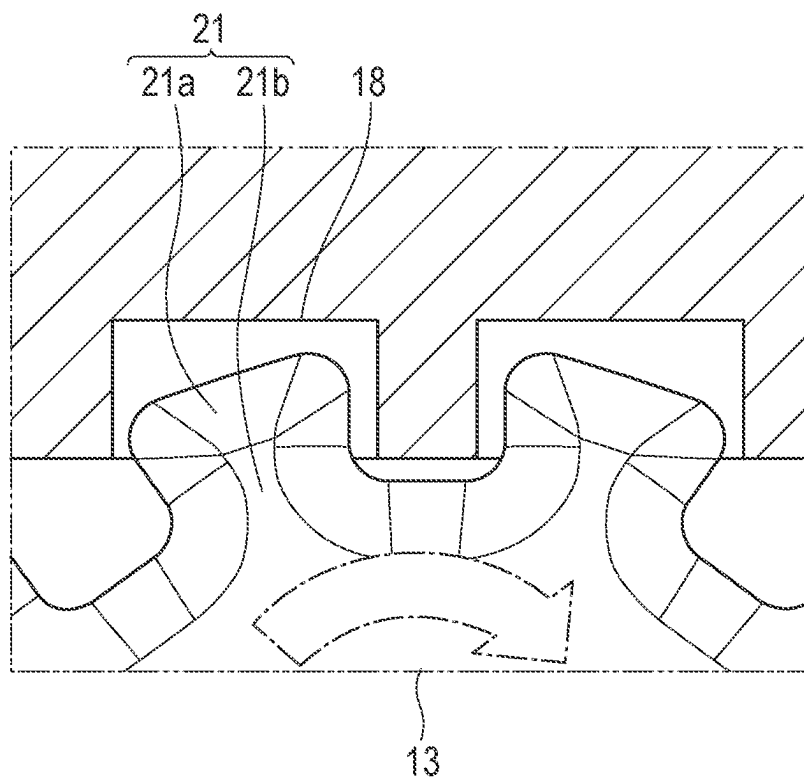


FIG. 7B

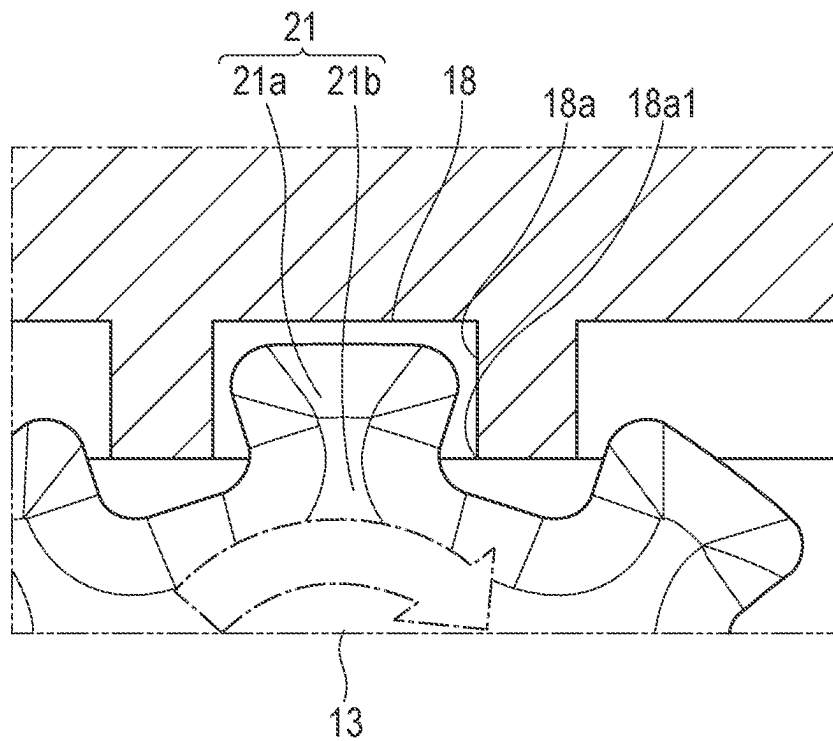


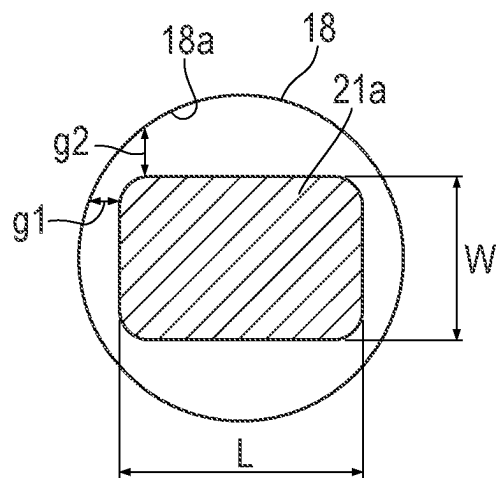
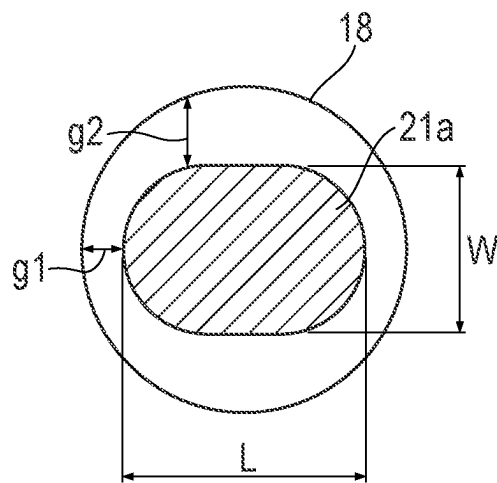
FIG. 8

FIG. 9

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LINEAR MOTION GUIDE APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is the U.S. national phase of International Application No. PCT/JP2021/040692 filed Nov. 5, 2021, which designated the U.S. and claims priority to JP 2021-012649 filed Jan. 29, 2021, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

The present invention relates to a linear motion guide apparatus including a mechanism that controls positional displacement of a cage.

BACKGROUND ART

A linear motion guide apparatus including a positional displacement control mechanism that controls positional displacement of a cage is known (refer to Patent Literature 1). The linear motion guide apparatus includes two guide rails that can move relative to each other in a longitudinal direction thereof via at least one rolling element. The rolling element is held by a cage.

When the two guide rails are repeatedly moved relative to each other in the linear motion guide apparatus, the cage that holds the rolling element is displaced from the normal position. The positional displacement control mechanism is provided to control positional displacement of the cage. The positional displacement control mechanism includes racks provided to the guide rails, and a pinion provided to the cage. When the two guide rails move relative to each other, the pinion moves in the normal position while meshing with the racks. Hence, it is possible to control positional displacement of the cage.

In the linear motion guide apparatus described in Patent Literature 1, the racks and the pinion are formed as involute gears that are ideal gears. Consequently, the strength of the racks and the pinion can be secured.

CITATION LIST**Patent Literature**

Patent Literature 1: JP S62-179423 U

SUMMARY OF INVENTION**Technical Problem**

However, the involute gears have a complicated shape, and there is a problem that it takes great time and effort to provide the guide rails with the involute gears.

The present invention has been made in view of the above problem, and an object thereof is to provide a linear motion guide apparatus in which a guide rail can be readily provided with a meshing portion that engages with a gear.

Solution to Problem

In order to solve the above problem, one aspect of the present invention is a linear motion guide apparatus including: two guide rails that are movable relative to each other in a longitudinal direction thereof via at least one rolling element; and a cage configured to hold the rolling element

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and rotatably hold a gear engaging with a meshing portion provided to at least one of the two guide rails, in which the meshing portion includes a plurality of circular holes arranged in the longitudinal direction on the guide rail, and each of the circular holes has a wall surface formed in a cylindrical shape.

Advantageous Effects of Invention

According to the present invention, the meshing portion that engages with the gear is formed as the plurality of circular holes. Therefore, the meshing portion can be easily provided to the guide rail.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view of a linear motion guide apparatus of one embodiment of the present invention.

FIG. 2 is an internal structural drawing of the linear motion guide apparatus of the embodiment.

FIG. 3 is a perspective view illustrating rolling elements and a gear of the linear motion guide apparatus of the embodiment.

FIG. 4 is a perspective view illustrating guide rails and the gear of the linear motion guide apparatus of the embodiment (the perspective view partially including a cross-sectional view of the guide rail).

FIGS. 5A and 5B are detailed views (FIG. 5A is a front view, and FIG. 5B is a cross-sectional view taken along line b-b of FIG. 5A) of the guide rail of the linear motion guide apparatus of the embodiment.

FIGS. 6A to 6C are detailed views (FIG. 6A is a perspective view, FIG. 6B is a side view, and FIG. 6C is a plan view) of the gear of the linear motion guide apparatus of the embodiment.

FIGS. 7A and 7B are diagrams illustrating circular holes and teeth of the gear of the linear motion guide apparatus of the embodiment (FIG. 7A illustrates a state in which the tooth of the gear starts to enter the circular hole, and FIG. 7B illustrates a state in which a tooth tip portion of the tooth is in contact with a wall surface of the circular hole).

FIG. 8 is a cross-sectional view of the circular hole and the tooth tip portion of the linear motion guide apparatus of the embodiment.

FIG. 9 is a cross-sectional view illustrating another example of the tooth tip portion.

DESCRIPTION OF EMBODIMENTS

Embodiments of a linear motion guide apparatus of the present invention are described hereinafter with reference to the accompanying drawings. However, the linear motion guide apparatus of the present invention can be embodied in various forms and is not limited to the embodiments described in the description. The embodiments are provided with the intention of enabling those skilled in the art to fully understand the invention by fully disclosing the description.

FIG. 1 is a perspective view of a linear motion guide apparatus 1 of one embodiment of the present invention. Reference signs 2 and 3 denote guide rails. Raceway grooves 4 and 5 having a V shape in cross section are formed in the guide rails 2 and 3, respectively. The raceway groove 4 includes raceway surfaces 7a and 7b perpendicular to each other. The raceway groove 5 includes raceway surfaces 8a and 8b perpendicular to each other. The guide rails 2 and 3 can move relative to each other in a longitudinal direction thereof via rollers 9a and 9b (refer to FIG. 2).

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A mounting hole **2a** for mounting on, for example, a base is formed in the guide rail **2**. A mounting hole **3a** for mounting on, for example, a table is formed in the guide rail **3**. The shape of the guide rail **3** is substantially the same as a shape of the guide rail **2** flipped 180°.

FIG. **2** illustrates a state in which the guide rail **3** on the front side in FIG. **1** is omitted. FIG. **3** illustrates a state in which a cage **10** of FIG. **2** is omitted. The rollers **9a** and **9b** as rolling elements are placed between the opposing raceway grooves **4** and **5** of the guide rails **2** and **3**. The rollers **9a** and **9b** serve as a crossed roller having the adjacent rollers **9a** and **9b** whose axes are perpendicular to each other. In side view, the rollers **9a** and **9b** have an approximately square shape, and the diameter of the rollers **9a** and **9b** is slightly greater than the length of the rollers **9a** and **9b** in the axial direction. The side surface of the roller **9a** is in contact with the raceway surface **7a** of the guide rail **2** and the raceway surface **8b** of the guide rail **3**. The side surface of the roller **9b** is in contact with the raceway surface **7b** of the guide rail **2** and the raceway surface **8a** of the guide rail **3**. A force acting on the guide rail **2** is transmitted to the guide rail **3** via the rollers **9a** and **9b**.

Note that in the embodiment, a crossed roller is used as the rolling element. However, a ball may be used as the rolling element, or a parallel roller such as a needle roller may be used as the rolling element.

As illustrated in FIG. **2**, the rollers **9a** and **9b** are held by the cage **10** extending in the longitudinal direction along the guide rails **2** and **3**. A pinion-like gear **13** is also rotatably held by the cage **10**. The gear **13** is placed substantially in the middle of the cage **10** in a length direction thereof.

FIG. **4** illustrates the guide rails **2** and **3** and the gear **13**. Undercuts **16** and **17** are formed in the bottoms of the raceway grooves **4** and **5** of the guide rails **2** and **3**. Rack-like meshing portions **2b** and **3b** are formed in the bottom surfaces of the undercuts **16** and **17**. Teeth **21** of the gear **13** engage with the meshing portions **2b** and **3b**. The position of the gear **13** relative to the guide rails **2** and **3** is determined by the relative positions of the guide rails **2** and **3**.

FIGS. **5A** and **5B** illustrate detailed views of the guide rail **2**. The meshing portion **2b** includes a plurality of circular holes **18** arranged in the longitudinal direction on the guide rail **2**. The circular holes **18** are evenly spaced in the longitudinal direction on the guide rail **2**. A wall surface **18a** of each of the circular holes **18** is formed in a cylindrical shape. In other words, the shape of the circular holes **18** in cross section is circular, and the shape of the circular holes **18** in cross section is uniform in a depth direction thereof. A center line **18c** of each of the circular holes **18** is orthogonal to the longitudinal direction of the guide rail **2**. As in the guide rail **2**, a plurality of the circular holes **18** is also formed in the guide rail **3** (refer to FIG. **4**).

The circular holes **18** are formed directly in the guide rails **2** and **3**. Moreover, the circular holes **18** are drilled by use of, for example, a cutting tool such as a drill or end mill having a flat tip. Furthermore, for example, a bottom surface **18b** of each of the circular holes **18** is formed flat (refer to FIG. **5B**). Note that the rack having the circular holes may be mounted on the guide rail, the circular holes may be drilled by, for example, electrical discharge machining, or the bottom surface of the circular hole may be formed in a conical shape adjusted to a drill having a conical tip.

FIGS. **6A** to **6C** illustrate detailed views of the gear **13**. As illustrated in FIG. **6A**, the gear **13** includes a disk portion **13a** having the many teeth **21** formed on the circumference, and a shaft portion **13b** formed integrally with the disk

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portion **13a**. The gear **13** is supported by the cage **10** in such a manner as to be rotatable about the shaft portion **13b**.

Each of the teeth **21** of the gear **13** includes a tooth tip portion **21a** and a dedendum portion **21b**. The tooth tip portion **21a** forms a shape obtained by rounding corners of a rectangular parallelepiped. As illustrated in FIGS. **6B** and **6C**, a length **L** of the tooth tip portion **21a** in a rotational direction **R** is greater than a face width **W** (a length in a rotation axis direction) of the tooth tip portion **21a**. Note that in FIG. **6C**, the tooth tip portion **21a** is hatched.

As illustrated in FIG. **6B**, as viewed in the rotation axis direction, the dedendum portion **21b** is constricted, and a length **T** of the dedendum portion **21b** in the rotational direction **R** is less than the length **L** of the tooth tip portion **21a** in the rotational direction **R**. FIGS. **7A** and **7B** illustrate the circular holes **18** and the teeth **21** of the gear **13**. FIG. **7A** illustrates a state in which the tooth **21** of the gear **13** starts to enter the circular hole **18**, and FIG. **7B** illustrates a state in which the tooth tip portion **21a** is in contact with the wall surface **18a** of the circular hole **18**. Note that in FIG. **7B**, it appears that there is a gap between the tooth tip portion **21a** and the wall surface **18a** of the circular hole **18**. However, since the tooth tip portion **21a** has the face width **W** in a direction into the page (a direction orthogonal to the page), the back side of the tooth tip portion **21a** is in contact with the wall surface **18a** of the circular hole **18** in the direction into the page. As illustrated in FIGS. **7A** and **7B**, the dedendum portion **21b** of the tooth **21** is constricted in such a manner as to avoid interference with a mouth portion **18a** of the circular hole **18**.

FIG. **8** illustrates a cross-sectional view of the circular hole **18** and the tooth tip portion **21a** (a cross-sectional view orthogonal to the axis of the circular hole **18**). As described above, the length **L** of the tooth tip portion **21a** in the rotational direction is greater than the face width **W** of the tooth tip portion **21a**. Hence, a gap **g1** in the rotational direction between the tooth tip portion **21a** and the wall surface **18a** of the circular hole **18** is smaller than a gap **g2** in the rotation axis direction. The gap **g2** is provided in such a manner as to permit axial runout of the gear **13** (the tilt of the gear **13**).

When the length **L** of the tooth tip portion **21a** in the rotational direction is greater than the face width **W** of the tooth tip portion **21a**, then the gap **g2** is larger than the gap **g1**. Hence, the cross-sectional area of the tooth tip portion **21a** can be increased while permitting axial runout of the gear **13**, and the strength of the tooth **21** can be increased. Moreover, the gap **g1** is smaller than the gap **g2**. Therefore, it is possible to avoid the gear **13** from moving in the rotational direction (the longitudinal direction of the guide rails **2** and **3**).

FIG. **9** illustrates another example of the cross-sectional shape of the tooth tip portion **21a**. As illustrated in FIG. **9**, the cross-sectional shape of the tooth tip portion **21a** may be a shape obtained by combining a rectangle and semicircles. Moreover, although not illustrated, the cross-sectional shape of the tooth tip portion **21a** may be elliptical.

The configuration of the linear motion guide apparatus **1** of the embodiment is described above. The linear motion guide apparatus **1** of the embodiment has the following effects:

Each of the meshing portions **2b** and **3b** includes the plurality of circular holes **18** arranged in the longitudinal direction on the respective guide rail **2** or **3**, and the wall surface **18a** of each of the circular holes **18** is formed in a cylindrical shape. Therefore, the shape of the meshing portions **2b** and **3b** that engage with the gear **13** can be

simplified, and the meshing portions **2b** and **3b** can be easily provided to the guide rails **2** and **3**.

The length **L**, in the rotational direction, of the tooth tip portion **21a** of the each of the teeth **21** of the gear **13** is greater than the face width **W** of the tooth tip portion **21a**. Therefore, the gap **g2** can be made larger than the gap **g1**, and the cross-sectional area of the tooth tip portion **21a** can be increased while permitting axial runout of the gear **13**. Moreover, the gap **g1** is smaller than the gap **g2**. Therefore, it is possible to avoid the gear **13** from moving in the rotational direction.

The dedendum portion **21b** of the each of the teeth **21** is constricted as viewed in the rotation axis direction of the gear **13**. Therefore, interference between the mouth portions **18a** of the circular holes **18** and the teeth **21** can be avoided.

The circular holes **18** are formed directly in the guide rails **2** and **3**. Therefore, the meshing portions **2b** and **3b** can be easily formed in the guide rails **2** and **3**.

The bottom surfaces **18b** of the circular holes **18** are formed flat. Therefore, it is possible to avoid interference between the circular holes **18** and the mounting holes **2a** and **3a** of the guide rails **2** and **3**.

The circular holes **18** are drilled in the guide rails **2** and **3** with a cutting tool. Therefore, the circular holes **18** can be easily machined in the guide rails **2** and **3**.

Note that the linear motion guide apparatus of the embodiment can be incorporated as an elementary component in a wide range of fields from industrial machine fields such as machine tools, semiconductor and liquid crystal manufacturing apparatuses (for example, mounters), and robots to consumer fields such as system kitchens, various game machines, medical machines, food machines, and conveyance apparatuses.

The present description is based on Japanese Patent Application No. 2021-012649 filed on Jan. 29, 2021. The entire contents thereof are included herein.

REFERENCE SIGNS LIST

- 1 Linear motion guide apparatus
- 2, 3 Guide rail
- 2b, 3b Meshing portion
- 9a, 9b Roller (rolling element)
- 10 Cage
- 13 Gear
- 18 Circular hole
- 18a Wall surface of circular hole
- 18b Bottom surface of circular hole
- 21 Tooth of gear
- 21a Tooth tip portion
- 21b Dedendum portion
- L Length of tooth tip portion in rotational direction
- W Face width of tooth tip portion
- The invention claimed is:
- 1. A linear motion guide apparatus comprising:
two guide rails that are movable relative to each other in a longitudinal direction thereof via at least one rolling element; and
a cage configured to hold the rolling element and rotatably hold a gear engaging with a meshing portion provided to at least one of the two guide rails, wherein the meshing portion includes a plurality of circular holes arranged in the longitudinal direction on the guide rail,

each of the circular holes has a wall surface formed in a cylindrical shape, and

a gap (**g1**) in a rotational direction of the gear between a tooth tip portion of each tooth of the gear and the wall surface of the circular hole is smaller than a gap (**g2**) in a rotation axis direction of the gear.

2. The linear motion guide apparatus according to claim 1, wherein the circular holes are formed directly in the guide rail.

3. The linear motion guide apparatus according to claim 1, wherein bottom surfaces of the circular holes are formed flat.

4. The linear motion guide apparatus according to claim 2, wherein bottom surfaces of the circular holes are formed flat.

5. A linear motion guide apparatus comprising:
two guide rails that are movable relative to each other in a longitudinal direction thereof via at least one rolling element; and

a cage configured to hold the rolling element and rotatably hold a gear engaging with a meshing portion provided to at least one of the two guide rails, wherein the meshing portion includes a plurality of circular holes arranged in the longitudinal direction on the guide rail, each of the circular holes has a wall surface formed in a cylindrical shape, and

a length, in a rotational direction, of a tooth tip portion of each tooth of the gear is greater than a face width of the tooth tip portion.

6. The linear motion guide apparatus according to claim 5, wherein the circular holes are formed directly in the guide rail.

7. The linear motion guide apparatus according to claim 6, wherein bottom surfaces of the circular holes are formed flat.

8. The linear motion guide apparatus according to claim 5, wherein bottom surfaces of the circular holes are formed flat.

9. A linear motion guide apparatus comprising:
two guide rails that are movable relative to each other in a longitudinal direction thereof via at least one rolling element; and

a cage configured to hold the rolling element and rotatably hold a gear engaging with a meshing portion provided to at least one of the two guide rails, wherein the meshing portion includes a plurality of circular holes arranged in the longitudinal direction on the guide rail, each of the circular holes has a wall surface formed in a cylindrical shape, and

a dedendum portion of each tooth of the gear is constricted as viewed in a rotation axis direction of the gear.

10. The linear motion guide apparatus according to claim 9, wherein the circular holes are formed directly in the guide rail.

11. The linear motion guide apparatus according to claim 10, wherein bottom surfaces of the circular holes are formed flat.

12. The linear motion guide apparatus according to claim 9, wherein bottom surfaces of the circular holes are formed flat.